Master's Thesis

Improving video prediction with SAM3 object-centric video representation

Hanbom Woo

Department of Electrical and Computer Engineering

The Graduate School

Sungkyunkwan University

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Supervised by
Il Yong Chun
Major Advisor

This certifies that the Master's Thesis
of Hanbom Woo is approved

심사위원장 signature

Committee Chair: Byeungwoo Jeon

심사위원 1 signature

Committee Member 1: Eun-Seok Ryu

지도교수 signature

Major Advisor: Il Yong Chun

The Graduate School
Sungkyunkwan University
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Abstract

Improving video prediction with SAM3 object-centric video representation

Video prediction aims to predict future frames from past observations by modeling temporal dynamics. Recent diffusion-based VP methods have achieved promising results in future frame generation. To provide explicit guidance for uncertain future dynamics, existing methods often condition the generation process on predicted future cues, such as motion or object representations. Object representations provide structured guidance by describing videos at the object level. However, if the predicted future object representations fail to maintain consistent object identities across frames, they still struggle to prevent over-smoothed object boundaries in predicted videos. Moreover, the fixed conditioning strategy used in diffusion-based video prediction can repeatedly propagate errors in these predicted cues as sampling bias throughout the generation process.

To address this, we propose **SAM3Future**, a conditional diffusion framework that leverages identity-consistent object video representations from Segment Anything 3 (SAM3) for future video prediction. SAM3Future predicts future SAM3 object embeddings by modeling object-wise temporal dynamics and uses them to construct object-centric video representations. To mitigate the impact of imperfect future predictions, we introduce a step-wise conditioning mechanism that supervises object-wise correlation maps with SAM3-derived masks, encouraging each object condition to be applied to its corresponding object region during denoising. Experiments on three video prediction benchmarks demonstrate that SAM3Future outperforms existing state-of-the-art VP methods.

Keywords: Video prediction, Conditional diffusion model, Object-centric representation, Step-wise guidance

1. Introduction

video prediction is a fundamental task in computer vision, with applications in decision-making systems[1, 2, 3], autonomous driving[4, 5, 6] and robotic navigation[7, 8, 9]. Its goal is to predict future red-green-blue (RGB) frames from past ones by modeling temporal dynamics. However, predicting future frames remains extremely challenging due to the inherent uncertainty in future dynamics.

Recent diffusion-based video prediction models[10, 11, 12, 13] have shown promising performance by modeling the stochastic distribution of future frames. To reduce future uncertainty while improving visual coherence with the observed past, diffusion-based video prediction models guide the generation process using predicted future cue that explicitly capture temporal dynamics [12, 13, 14]. These cues typically fall into two categories: motion representations and object representations. Specifically, object representations provide structured inductive bias for video prediction by decomposing a video into an object-level component, allowing future dynamics to be modeled as transitions of individual object states. Existing object-centric video prediction methods, therefore, predict future object representations and use them to condition the generation process.

However, existing object-centric video prediction methods [15, 14, 16] commonly rely on slot-based representations. Since such unsupervised slots are not explicitly associated with persistent object identities or instance-level masks, they mainly serve as object-level token conditions. Therefore, slot-conditioning mechanisms do not enforce spatial alignment between each object condition and its corresponding object region. This weak localization of object conditions can make it difficult to prevent object-background mixing and over-smoothed object boundaries. Moreover, existing conditional diffusion video

prediction models use fixed conditioning cues, allowing errors in the pseudo cue to be repeatedly propagated as sampling bias during generation, as illustrated in Figure 1 (a).

Instead, we leverage the unified vision foundation model Segment Anything 3 (SAM3) [17] to obtain object representations that are explicitly associated with object instances. Unlike unsupervised slots, SAM3-derived object embeddings are obtained together with instance-level masks and are designed to distinguish different object instances while maintaining their identities across frames.

Based on this property, we propose **SAM3Future**, the first conditional diffusion video prediction framework that exploits the object representations of past frames from the SAM3 to generation-adaptive future object-centric guidance. SAM3Future predicts future object representations and constructs object-centric conditions through segmentation-supervised spatial alignment during the denoising process. Rather than relying on fixed pseudo future cues throughout sampling, SAM3Future updates the object-centric condition in a denoising-step-adaptive manner. To this end, we propose two key modules that predict and update object-centric video representations for future frames. First, The **Future Object Predictor** projects SAM3 object embeddings to the Diffusion model and then predicts future object embeddings by modeling object-wise temporal dynamics. Second, the **Step-Wise Spatial Conditioning** module spatially localizes the predicted future object embeddings on intermediate denoising features, where the resulting localization maps are supervised with SAM3-derived masks.

Our contributions are summarized as follows:

1. We propose **SAM3Future**, a new conditional diffusion video prediction framework that leverages SAM3 to obtain object representations for future prediction by exploiting instance-level object embeddings and their corresponding mask.

2. We introduce the **Future Object Predictor**, which predicts future object embeddings from past SAM3 object embeddings by modeling object-wise temporal dynamics.

3. We introduce **Step-Wise Spatial Conditioning** module, a segmentation-supervised spatial conditioning mechanism that aligns predicted object embeddings with intermediate diffusion features during the denoising process.

4. Experiments on three video prediction benchmarks show that SAM3Future outperforms several state-of-the-art(SOTA) video prediction methods.

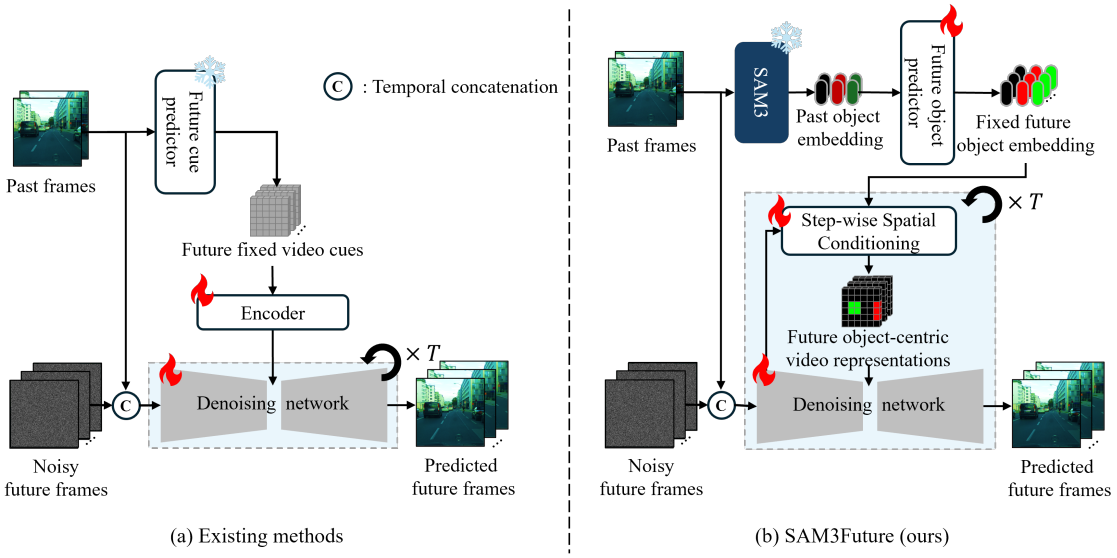


Figure 1: Overview of the proposed SAM3Future framework vs. existing diffusion-based video prediction approaches with predicted future condition. (a) the existing approaches use a fixed predicted future condition throughout the generation process. (b) The proposed SAM3Future framework first predicts future object embeddings from SAM3-derived object representations and constructs step-wise updated condition map, enabling object embeddings to be selectively localized at each denoising step.

2. Related Works

2.1. Video prediction

Early video prediction methods employed recurrent neural networks (RNN) [18, 19] which often produced blurry predictions due to deterministic future modeling. To address the inherent uncertainty of future dynamics in videos, subsequent works introduced stochastic latent variable models, such as VAEs and VRNNs[20, 21].

Recently, diffusion models[22, 10, 11] have been widely adopted for video prediction due to their ability to model complex stochastic distributions through iterative denoising. Residual Video Diffusion (RVD)[22] predicts stochastic residual changes between consecutive frames. Masked Conditional Video Diffusion (MCVD)[11] trains a conditional diffusion model with random masking over past frames, and extends long-term prediction through block-wise autoregressive sampling.

To provide more informative guidance for explicitly modeling future dynamics, recent methods leverage predicted pseudo future cues[12, 13], such as motion representations[12, 13] or object representations[14]. Distribution Extrapolation Diffusion Model (ExtDM)[12] extrapolates future motion cues from present frames and reconstructs future frames from the extrapolated cues. Spatio-Temporal Diffusion (STDiff)[13] decomposes videos into motion and content, predicts future motion features with a neural SDE[23], and conditions the diffusion model on the predicted motion.

While motion representations mainly describe dense low-level temporal changes, object representations provide a higher-level structure by modeling future dynamics as object-wise state transitions.

2.2. Object-centric video prediction

Object-centric video prediction methods [15, 24, 14, 16] use object representations as a structured inductive bias for video prediction by decomposing a video into object-level components. This decomposition allows future dynamics to be modeled as object-wise changes in appearance and position, enabling future frames to be synthesized from the predicted object representations. Existing methods commonly use slot-based object representations for this purpose. OCVP[15] parses frames into object slots, predicts future object states with an object-centric transformer, and renders future frames from the predicted slots. SlotDiffusion[14] incorporates slot-based object representations into diffusion-based generative modeling. LPWM[16] further represents scenes with latent particles and models stochastic object-centric dynamics for video prediction.

Despite their effectiveness, slot-based object-centric methods typically learn object representations through unsupervised clustering with reconstruction or denoising objectives. Therefore, each slot is not explicitly associated with a persistent semantic identity or an instance-level segmentation mask. If such object representations are used only as token-level conditions, the conditioning mechanism does not directly supervise whether each object condition is spatially aligned with its corresponding instance region. As a result, a predicted slot may drift to a different object or entangle multiple objects in complex video scenes, leading to inconsistent object-level guidance across frames. When such inconsistent pseudo cues are used as fixed conditions, their errors can be repeatedly injected into the denoising process, biasing the generation trajectory.

2.3. Segment Anything 3

SAM3 [17] is a unified vision foundation model that detects, segments, and tracks objects in images and videos based on concept prompts, such as text, bounding boxes, and

points. It leverages vision-language aligned Perception Encoder[25] features to produce semantics-aware object embeddings conditioned on the given prompts. Unlike slot-based object representations, SAM3 explicitly identifies object instances that match the given prompt and localizes them with segmentation masks. These instances are then tracked across video frames to preserve their identities. As a result, SAM3 object embeddings provide instance-discriminative and temporally consistent object representations for conditional video prediction.

Motivated by these properties, we predict future SAM3 object representations and use them as identity-consistent anchors for step-wise adaptive conditioning during diffusion sampling.

3. Background

Denoising diffusion probabilistic models (DDPMs)[26] are latent variable models defined by the the forward and reverse processes. Given a clean sample $\mathbf{x}_0$, the forward process is defined as a Markovian chain that gradually adds isotropic Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to $\mathbf{x}_0$ over T iterations. This process yields intermediate latent variables $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}, \quad t = 1, \ldots, T$, where α_t denotes a constant determined by the noise schedule at timestep t. The reverse process starts from Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and gradually denoises it using a denoising network $D_\theta(\mathbf{x}_t, t)$ parameterized by θ.

For conditional generation, the reverse process models the denoising trajectory conditioned on $\mathbf{y}$. We optimize the conditional denoiser $D_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y})$ with the noise prediction objective:

$$\mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}, t}\left[\|\boldsymbol{\epsilon} - D_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y})\|_2^2\right], \tag{1}$$

For the remainder of this paper, we will refer to equation (1) as the loss of video diffusion $\mathcal{L}_{\text{diff}}$.

4. Methods

The proposed SAM3Future framework predicts a set of future RGB frames $\mathbf{X} \in \mathbb{R}^{F \times H \times W \times 3}$, given a set of past RGB frames $\mathbf{Y} \in \mathbb{R}^{P \times H \times W \times 3}$, where H and W denote the spatial height and width, while F and P denote the temporal lengths of the future and past frames, respectively. Following a conditional diffusion paradigm, our framework iteratively denoise a set of pure Gaussian noise frames $\mathbf{Z} \in \mathbb{R}^{F \times H \times W \times 3}$ over T steps using a modified Hourglass Diffusion Transformer (HDiT)[27], resulting in a set of future predicted frames $\hat{\mathbf{X}} \in \mathbb{R}^{F \times H \times W \times 3}$

A primary goal of our framework is to leverage strong object modeling capability of SAM3 to serve as effective condition for the generation process. To achieve this, the proposed Future Object Predictor and Step-Wise Spatial Conditioning module construct an object-centric future video representations from SAM3 embeddings and iteratively update these representations. We use these representations to condition the generation process, allowing the model to receive step-specific guidance only where accurate. The overall architecture of SAM3Future is illustrated in Figure 2.

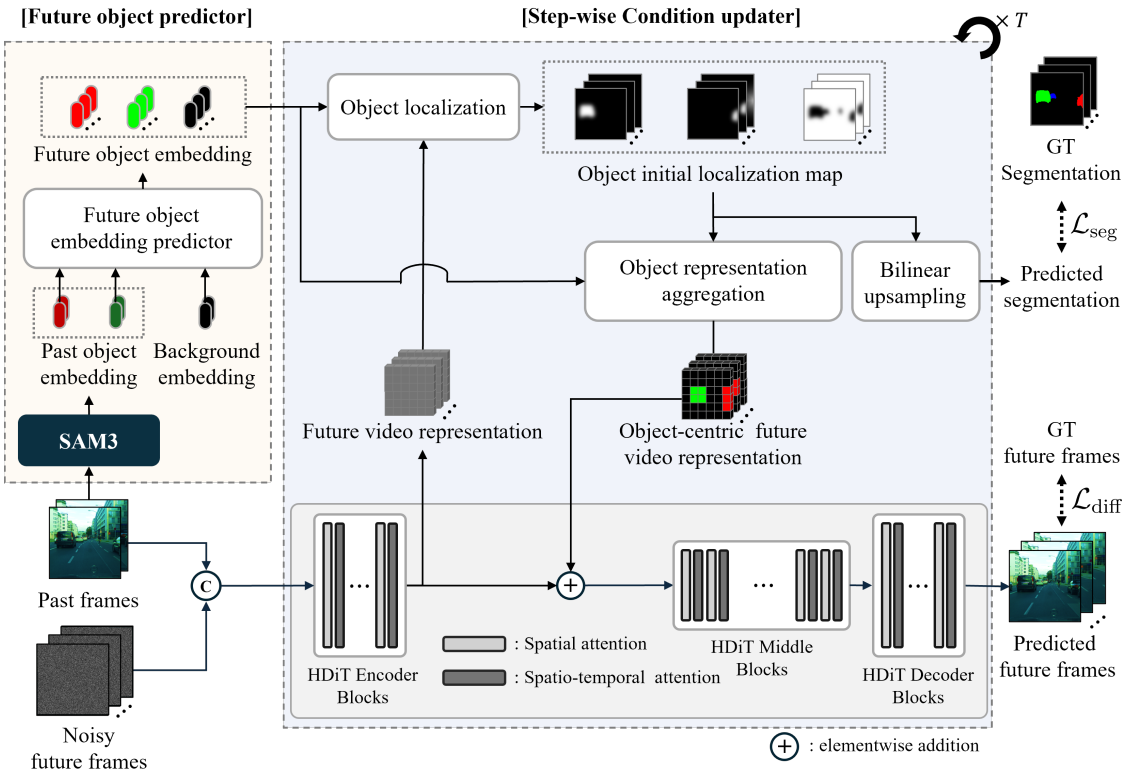


Figure 2: The overall architecture of the proposed SAM3Future video prediction diffusion model in the reverse diffusion process. Given past frames $\mathbf{Y}$ and predicted future object embeddings $\mathbf{C}$, SAM3Future generates future frames by iteratively denoising pure noise $\mathbf{Z}$ over T diffusion steps. The Future Object Predictor estimates future object embeddings $\hat{\mathbf{e}}$ from SAM3 past object embeddings, while the Step-wise Conditioning Module localizes these embeddings on noisy future frames to construct step-adaptive conditions $\mathbf{c}$ based on object-centric video representations.

4.1. Preliminaries

Conditional diffusion for video prediction

In video prediction, the clean sample X_0 corresponds to the target future frames to be generated. Conditional diffusion models generate X_0 from observed past frames Y and additional future representations C. The conditional distribution over X_0 can be written by marginalizing the reverse diffusion trajectory as

$$p_\theta(\mathrm{X}_0|\mathrm{Y},\mathrm{C}) = \int p(\mathrm{X}_T) \prod_{t=1}^{T} p_\theta(\mathrm{X}_{t-1}|\mathrm{X}_t,\mathrm{Y},\mathrm{C}_t) d\mathrm{X}_{1:T} \tag{2}$$

In existing approaches, future representations are typically predicted once by an existing predictor $\mathcal{F}_\phi$, such as $\mathrm{C}_{\text{fixed}} = \mathcal{F}_\phi(\mathrm{Y})$, and are then treated as a fixed observation throughout all denoising steps, i.e., $\mathrm{C}_t = \mathrm{C}_{\text{fixed}}$ for all t. Then, conditional denoising process can be interpreted as approximate the score $\nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{X}_t|\mathrm{Y},\mathrm{C}_{\text{fixed}})$. By Bayes rule, this score can be decomposed as

$$\nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{X}_t|\mathrm{Y},\mathrm{C}_{\text{fixed}}) = \nabla_{\mathrm{X}_t} \log \left(\frac{p_\theta(\mathrm{X}_t) p_\theta(\mathrm{Y}|\mathrm{X}_t) p_\theta(\mathrm{C}_{\text{fixed}}|\mathrm{X}_t,\mathrm{Y})}{p_\theta(\mathrm{Y},\mathrm{C}_{\text{fixed}})} \right) \tag{3}$$

$$= \nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{X}_t) + \nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{Y}|\mathrm{X}_t) + \nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{C}_{\text{fixed}}|\mathrm{X}_t,\mathrm{Y}) \tag{4}$$

Here, $p_\theta(\mathrm{C}_{\text{fixed}}|\mathrm{X}_t,\mathrm{Y})$ measures how well the fixed future cue is aligned with the current denoising state X_t under the observed past frames Y.

Step-wise Conditional diffusion for video prediction

However, in video prediction, the future cue is a pseudo cue estimated from the past and thus inevitably contains prediction errors. We therefore replace the fixed cue with a step-wise updated condition C_t. Specifically, given an initial cue $\mathrm{C}_{\text{fixed}}$, we compute a step-wise condition as $\mathrm{C}_t = \mathcal{F}_\theta(\mathrm{X}_t, \mathrm{C}_{\text{fixed}})$. The corresponding conditional score can be written as

$$\nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{X}_t|\mathrm{Y}, \mathrm{C}_t) = \nabla_{\mathrm{X}_t} \log \left(\frac{p_\theta(\mathrm{X}_t) p_\theta(\mathrm{Y}|\mathrm{X}_t) p_\theta(\mathrm{C}_t|\mathrm{X}_t, \mathrm{Y})}{p_\theta(\mathrm{Y}) p_\theta(\mathrm{C}_t|\mathrm{Y})} \right) \tag{5}$$

$$= \nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{X}_t) + \nabla_{\mathrm{X}_t} \log p_\theta(\mathrm{Y}|\mathrm{X}_t) + \nabla_{\mathrm{X}_t} \log \frac{p_\theta(\mathrm{C}_t|\mathrm{X}_t, \mathrm{Y})}{p_\theta(\mathrm{C}_t|\mathrm{Y})} \tag{6}$$

Unlike fixed conditioning, the step-wise condition score is normalized by $p_\theta(\mathrm{C}_t|\mathrm{Y})$. This normalization compares how well the step-wise condition C_t is explained by the current denoising state X_t and the past observation Y, relative to how well it can already be explained by Y alone. This perspective motivates our step-wise condition updater, which constructs future object-centric video representations by selectively localizing valid object embeddings at each step.

4.2. Future Object Predictor

In this module, we extract past object embeddings via SAM3 and predict future object embeddings for video prediction. Since we aim to leverage these representations as effective conditioning signals for video prediction rather than simply replicating them, we train the predictor to generate the corresponding future object embeddings for video prediction.

Extracting SAM3 object embeddings from past frames

We first extract object embeddings from the past frame set $\mathbf{Y}$ with pretrained SAM3 model. Since SAM3 requires prompts—such as bounding boxes, points, or text—to extract object representations, we employ a YOLOv26[28] detector to automatically generate bounding boxes for detecting up to K objects. Based on these bounding boxes, we extract past object embeddings $\mathbf{e} \in \mathbb{R}^{P\times K\times D}$, where D denotes the feature dimension of the SAM3 object embeddings.

Next, we project the SAM3 object embeddings $\mathbf{e}$ into the channel dimension of the diffusion representations. Specifically, we transform the embeddings into projected object embeddings $\mathbf{e}^{+} \in \mathbb{R}^{P\times K\times C}$, where C denotes the channel dimension of the diffusion representations, by applying a linear projection.

Meanwhile, to handle background regions not defined as objects, we introduce an additional learnable query $\mathbf{b} \in \mathbb{R}^{P\times 1\times C}$ that serves as a background embedding. By concatenating the projected object embeddings $\mathbf{e}^{+}$ and the background embedding $\mathbf{b}$ along the object dimension, we obtain the augmented embeddings $\tilde{\mathbf{e}} \in \mathbb{R}^{P\times (K+1)\times C}$. We then predict future embeddings from $\tilde{\mathbf{e}}$ to construct object-centric video representations for future frames.

Predicting future object embeddings for video prediction

These augmented embeddings $\tilde{\mathbf{e}}$ are then fed into a future object embedding predictor $\mathcal{P}$. We design $\mathcal{P}$ as a multi-stage transformer architecture to flexibly accommodate different numbers of past frames P and future frames F across experimental setups. The predictor $\mathcal{P}$ consists of L sequential stages $\{\mathcal{P}^{(l)} : l = 1, \ldots, L\}$, where $\mathcal{P}^{(l)}$ denotes the lth stage. At the first stage, $\mathcal{P}^{(1)}$ takes $\tilde{\mathbf{e}}_{1:P}$ as input and predicts object embeddings for frames $P+1$ to $2P$. For $l = 2, \ldots, L$, each stage $\mathcal{P}^{(l)}$ takes the current embedding sequence $\hat{\mathbf{e}}_{1:2^{l-1}P}$ as input and predicts object embeddings for frames $2^{l-1}P + 1$ to $2^l P$. The predicted embeddings are then appended to the input sequence, progressively extending its temporal length:

$$\hat{\mathbf{e}}_{1:2P} = \left(\tilde{\mathbf{e}}_{1:P}, \mathcal{P}^{(1)}\left(\tilde{\mathbf{e}}_{1:P}\right)\right) \tag{7}$$

$$\hat{\mathbf{e}}_{1:2^l P} = \left(\hat{\mathbf{e}}_{1:2^{l-1}P}, \mathcal{P}^{(l)}\left(\hat{\mathbf{e}}_{1:2^{l-1}P}\right)\right), \quad l = 2, \ldots, L. \tag{8}$$

After L stages, this process yields an extended object embedding sequence $\hat{\mathbf{e}}_{1:2^L P} \in \mathbb{R}^{(2^L P)\times(K+1)\times C}$. From this sequence, we take the embeddings corresponding to frames $P+1$ to $P+F$ as the predicted future object embeddings $\hat{\mathbf{e}} \in \mathbb{R}^{F\times(K+1)\times C}$.

Each stage $\mathcal{P}^{(l)}$ consists of M transformer blocks with object-wise temporal attention layers, which are designed to capture future semantic changes for each object, as illustrated in Figure 3. In each temporal attention layer, an object embedding is updated by aggregating information from the same object across frames. At the l-th stage, the input consists of object embeddings $(\mathbf{e}_1, \ldots, \mathbf{e}_N)$, where each object trajectory has length $N = 2^{l-1}P$. These embeddings are rearranged into temporal sequences $(\mathbf{e}^1, \ldots, \mathbf{e}^{K+1})$, where $\mathbf{e}^k = (\mathbf{e}^k_1, \ldots, \mathbf{e}^k_N) \in \mathbb{R}^{N\times C}$ denotes the temporal history of the k-th object.

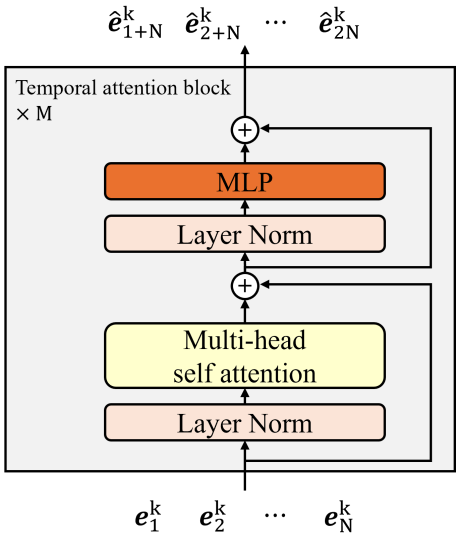


Figure 3: Temporal transformer block employed in the Future Object Predictor. The attention layer updates each object embedding along its temporal sequence without information across different objects.

For each object trajectory e^k, we obtain query, key, and value embeddings through learnable linear projections:

$$\mathrm{Q}^{(l),k} = \mathrm{e}^k \mathrm{W}_Q^{(l)}, \quad \mathrm{K}^{(l),k} = \mathrm{e}^k \mathrm{W}_K^{(l)}, \quad \mathrm{V}^{(l),k} = \mathrm{e}^k \mathrm{W}_V^{(l)}. \tag{9}$$

The object-wise temporal attention is then computed as

$$\mathrm{Attention}^{(l),k} = \mathrm{softmax}\left(\frac{\mathrm{Q}^{(l),k}\left(\mathrm{K}^{(l),k}\right)^\top}{\sqrt{C}}\right)\mathrm{V}^{(l),k}. \tag{10}$$

By applying this attention independently to each object trajectory, the predictor updates the embedding of each object according to its own temporal sequence. Through this object-wise temporal attention, $\mathcal{P}^{(l)}$ models object-specific future dynamics while preserving

temporal consistency within each object.

4.3. Modified HDiT architecture for video prediction

To predict future frames, we concatenate pure Gaussian noise frames $\mathbf{Z}$ with past frames $\mathbf{Y}$ as the initial denoising input to a modified HDiT [27]. The model iteratively denoises $\mathbf{Z}$ into the predicted future frames $\hat{\mathbf{X}}$.

The original HDiT architecture is designed for image generation consists with encoder, middle, and decoder DiT [29] blocks. Each block consists of neighborhood and global spatial attention layers, while the encoder and decoder incorporate downsampling and upsampling operations via pixel unshuffle and shuffle, respectively. We modified this architecture for video prediction by adding spatio-temporal attention layers after each spatial attention layers.

After processing the input through the encoder blocks, we obtain intermediate future video representations $\mathbf{r} \in \mathbb{R}^{F \times h \times w \times C}$, where h and w denote the height and width of intermediate future video representations. To effectively guide the video prediction process, we leverage the step-wise object-centric future video representations derived from Step-Wise Condition Updater as additional condition.

4.4. Step-wise Spatial Conditioning module

This module constructs step-wise future object-centric video representations by aligning the predicted object embedding $\hat{\mathbf{e}}$ with the current future denoising features. Specifically, we compute correlation maps between the embeddings and the denoising features, and supervise these maps with SAM-derived pseudo segmentation masks. This segmentation supervision encourages each object condition to be spatially aligned with its corresponding object region, rather than being used only as a weakly localized token-level condition. The aligned maps are then used to aggregate object embeddings into dense object-centric conditions for step-wise denoising.

Localization of object representations

To construct the object-centric video representations, we generate an object localized map $\mathbf{m} \in \mathbb{R}^{F\times(K+1)\times h\times w}$ through the correlation between the intermediate future video representations $\mathbf{r}$ and the predicted future object embeddings $\tilde{\mathbf{e}}$. We upsample the correlation map using a bilinear upsampler followed by a sigmoid activation to extract pixel-level instance segmentation probability maps $\hat{\mathbf{s}} \in \mathbb{R}^{F\times(K+1)\times H\times W}$. This segmentation probability maps $\hat{\mathbf{s}}$ are supervised using a segmentation loss against the pseudo-ground truth (GT) mask generated by SAM3.

$$\mathbf{m} = \langle \mathbf{r}, \hat{\mathbf{e}} \rangle, \quad \hat{\mathbf{s}} = \mathrm{Sigmoid}(\mathcal{U}(\mathbf{m})) \tag{11}$$

where $\langle \cdot, \cdot \rangle$, $\mathcal{U}(\cdot)$, and $\mathrm{Sigmoid}(\cdot) : \mathbb{R} \to [0, 1]$ denotes the correlation operation, bilinear interpolation upsampler, and the sigmoid function, respectively.

Object representations aggregation

Based on the future object embedding $\hat{\mathbf{e}}$ and the estimated object-wise segmentation map $\mathbf{m}$, we construct the step-wise updated future object-centric video representations $\mathbf{c} \in \mathbb{R}^{F \times H \times W \times C}$:

$$\mathbf{c} = \sum_{k=0}^{K+1} \text{Softmax}(\mathbf{m}_k) \cdot \hat{\mathbf{e}}_k \tag{12}$$

where Softmax$(\cdot)$ denotes the softmax function. Subsequently, to ensure stable optimization, $\mathbf{c}$ is linearly projected and integrated with the intermediate future video representations $\mathbf{r}$ through a residual connection, yielding the updated feature.

4.5. Train objective

To encourage accurate future-frame prediction and reliable step-wise object-centric conditioning, we additionally introduce a segmentation loss. The model is jointly trained with both the video diffusion loss $\mathcal{L}_{\text{diff}}$ and the segmentation loss $\mathcal{L}_{\text{seg}}$. The segmentation loss provides spatial alignment supervision for the object-wise correlation maps produced by the Step-Wise Spatial Conditioning module. This encourages each predicted object condition to be applied to its corresponding object region in the intermediate denoising features.

Given the predicted future segmentation map $\hat{\mathbf{s}}$ and the pseudo ground-truth segmentation map $\mathbf{s}$ obtained from SAM3, the overall training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}(\mathbf{s}, \hat{\mathbf{s}}), \tag{13}$$

where $\lambda_{\text{seg}} = 0.05$ balances the segmentation objective.

We define $\mathcal{L}_{\text{seg}}$ as a weighted sum of focal binary cross-entropy loss[30] and Dice

loss[31]:

$$\mathcal{L}_{\text{seg}}(\mathbf{s}, \hat{\mathbf{s}}) = \lambda_{\text{focal}}\mathcal{L}_{\text{focal}}(\mathbf{s}, \hat{\mathbf{s}}) + \lambda_{\text{dice}}\mathcal{L}_{\text{dice}}(\mathbf{s}, \hat{\mathbf{s}}). \tag{14}$$

Following common practice in instance segmentation models[32, 33], we set $\lambda_{\text{focal}} = 20.0$ and $\lambda_{\text{dice}} = 1.0$.

5. Experimental results and discussion

This section describes the experimental setups and presents results with some discussion. We compared our method with two categories of SOTA video prediction methods: (1) object-centric video prediction methods, (2) diffusion based video prediction methods. In addition, we investigate the contribution of different SAM3Future variants.

5.1. Experimental setup: datasets

We evaluate our method on two video prediction dataset Cityscapes[34], KTH[35] as well as one object-centric video prediction dataset Obj3D[36] that are widely used for both tasks. We briefly describe the datasets used in our experiments.

Cityscapes

The Cityscapes dataset provides RGB driving videos captured in urban street scenes. It consists of short video sequences with 30 frames. All frames are originally provided at a resolution of 1024×2048 pixels. For our experiments, each video is center-cropped and resized to 128×128. We use the first two frames as conditioning inputs and predict the remaining 28 frames.

KTH

The KTH human action dataset consists of videos of 25 subjects performing six action categories: running, jogging, walking, boxing, hand-clapping, and hand-waving. All frames are originally provided at a resolution of 320×240 pixels. For our experiments, we center-crop and resize them to 64×64. Following the experimental setup of MCVD[11], we use videos from 20 subjects for training and the remaining five subjects for testing. We

use 10 frames as conditioning inputs and predict the following 30 frames.

Obj3D

Obj3D is a simulated 3D dataset commonly used for object-centric video prediction. It features dense object interactions with deterministic dynamics. The dataset consists of 100-frame videos at a resolution of 128×128, where a randomly colored ball rolls toward multiple CLEVR-like[37] objects placed near the scene center and causes collisions. For our experiments, we use the first six frames as conditioning inputs and predict the following 44 frames.

5.2. Experimental setup: evaluation metrics

We evaluate our model using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM)[38], and Learned Perceptual Image Patch Similarity (LPIPS)[39]. For video prediction datasets, we additionally report Frechet Video Distance (FVD)[40], following previous experimental setups. For each test example, we generate 10 random predictions, instead of using more than 10 or 100 samples as in previous stochastic video prediction models.

Peak Signal-to-Noise Ratio (PSNR)

PSNR measures pixel-level reconstruction quality by comparing the maximum possible pixel value with the Mean Squared Error (MSE) between the ground-truth video frame $\mathbf{X}$ and the generated video frame $\hat{\mathbf{X}}$:

$$\text{MSE} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} [\mathbf{X}_(i,j) - \hat{\mathbf{X}}(i,j)]^2, \tag{15}$$

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_v^2}{\text{MSE}} \right), \tag{16}$$

where H and W denote the height and width of each video frame, respectively, and MAX_v is the maximum possible pixel value. A higher PSNR indicates lower pixel-level reconstruction error.

Structural Similarity Index Measure (SSIM)

SSIM measures the perceptual similarity between two video frames by considering luminance, contrast, and structural information:

$$\mathrm{SSIM}(\mathbf{X}, \hat{\mathbf{X}}) = l(\mathbf{X}, \hat{\mathbf{X}}) \cdot c(\mathbf{X}, \hat{\mathbf{X}}) \cdot s(\mathbf{X}, \hat{\mathbf{X}}), \tag{17}$$

where $l(\cdot,\cdot)$, $c(\cdot,\cdot)$, and $s(\cdot,\cdot)$ denote the luminance, contrast, and structure comparison terms, respectively. An SSIM value closer to 1 indicates higher structural similarity.

Learned Perceptual Image Patch Similarity (LPIPS)

LPIPS measures perceptual similarity between two video frames by computing the L2 distance between deep feature representations extracted from a pre-trained VGG [41]:

$$d(\mathbf{X}, \hat{\mathbf{X}}) = \sum_{l} \frac{1}{H_l W_l} \sum_{h,w} w_l \| \mathbf{f}^l(h,w) - \hat{\mathbf{f}}^l(h,w) \|_2^2, \tag{18}$$

where $\mathbf{f}^l(h,w)$ and $\hat{\mathbf{f}}^l(h,w)$ denote the feature vectors at spatial location (h,w) in the l-th layer, and w_l represents the learnable weight that scales the contribution of each layer. A lower LPIPS score indicates higher perceptual similarity.

Frechet Video Distance (FVD)

FVD measures the visual quality and temporal coherence of generated videos by computing the Frechet distance, or Wasserstein-2 distance, between the feature distributions of real videos r and generated videos g. The features are extracted using

a pre-trained video recognition model, such as I3D [42]. A lower FVD indicates that the distribution of generated videos is closer to that of real videos. It is defined as:

$$\text{FVD} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|_2^2 + \text{Tr}(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_g - 2(\boldsymbol{\Sigma}_r\boldsymbol{\Sigma}_g)^{1/2}) \tag{19}$$

where $(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r)$ and $(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ denote the mean vectors and covariance matrices of the feature distributions of real and generated videos, respectively, and Tr denotes the trace of a matrix.

5.3. Experimental setup: implementation details

We use the modified HDiT architecture as the video diffusion backbone. The backbone consists of three neighborhood attention blocks and eight global attention blocks. For the Future object predictor, we employ a temporal Transformer with $M = 4$ blocks.

For object embedding extraction, we use YOLOv26[28] to detect bounding boxes from first past frames. We set the maximum number of objects to $K = 4$ and use the detected boxes as context prompts for SAM3[17] to extract past object embeddings.

In training the diffusion model, we follow the default setup described in [27] and use an exponential moving average of model weights with a decay factor of 0.995. For sampling, we use DDIM with $T = 100$ reverse steps.

For existing SOTA methods, we use the default setups specified in their corresponding papers. We train the model for 1000 epochs with a batch size of 32. All experiments are conducted on an NVIDIA GeForce RTX 4090 GPU.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	FVD ↓
OCVP[15]	19.89	0.587	0.338	877.8
SlotDiffusion[14]	20.03	0.603	0.311	824.6
LPWM[16]	20.13	0.607	0.313	741.8
RVD[22]	17.21	0.489	0.242	465.0
RaMViD[10]	13.14	0.454	0.395	812.6
VIDM[43]	18.49	0.539	0.252	724.7
LFDM[44]	20.32	0.579	0.157	194.9
VDT[45]	–	**0.880**	–	142.3
MCVD[11]	22.50	0.720	0.121	184.8
STDiff[13]	21.26	0.658	0.136	107.3
ExtDM[12]	<u>22.84</u>	0.745	<u>0.108</u>	121.3
SyncVP[46]	20.93	0.649	0.159	<u>84.0</u>
SAM3Future	**24.31**	<u>0.747</u>	**0.082**	**78.41**

Table 1: Quantitative comparison on Cityscapes dataset.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	FVD ↓
OCVP[15]	20.97	0.606	0.264	613.3
SlotDiffusion[14]	21.25	0.618	0.253	443.7
LPWM[16]	21.90	0.657	0.252	414.8
RVD[22]	25.32	0.782	0.128	441.1
RaMViD[10]	23.42	0.590	0.169	581.3
VIDM[43]	25.02	0.694	0.150	357.1
LFDM[44]	26.89	0.772	0.110	320.2
MCVD[11]	27.50	0.835	0.092	323.0
STDiff[13]	24.88	0.828	0.096	278.37
ExtDM[12]	<u>28.53</u>	<u>0.838</u>	<u>0.082</u>	<u>227.9</u>
SAM3Future	**28.79**	**0.856**	**0.054**	**139.6**

Table 2: Quantitative comparison on KTH dataset.

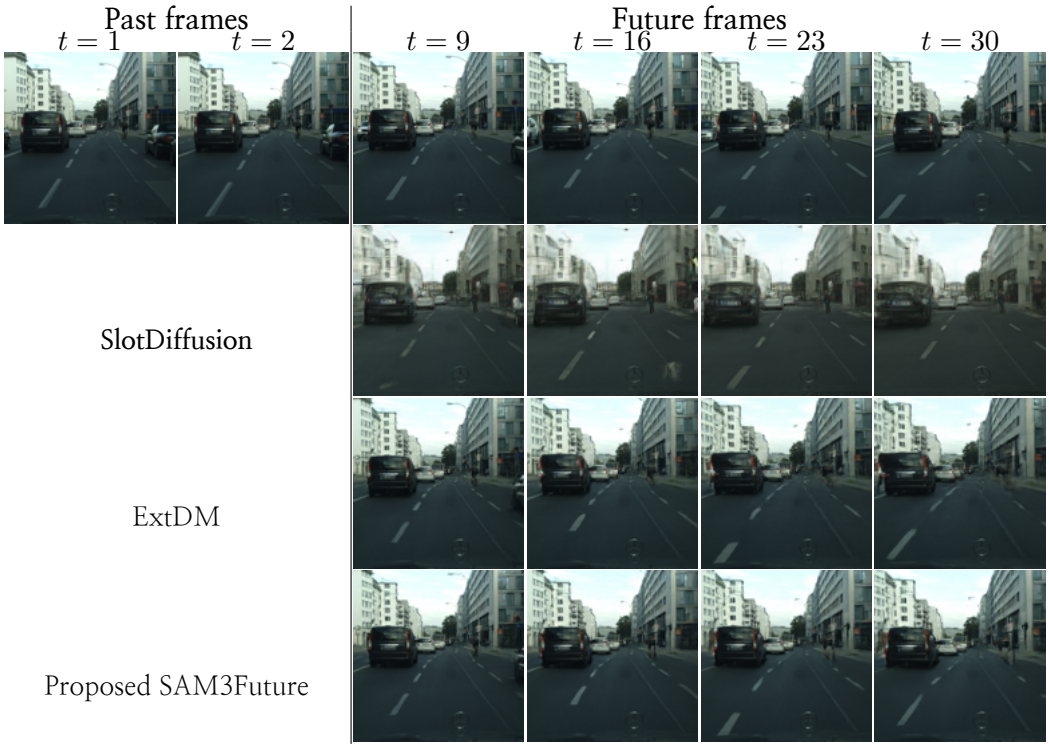


Figure 4: Qualitative comparisons between different video prediction models Cityscapes dataset.

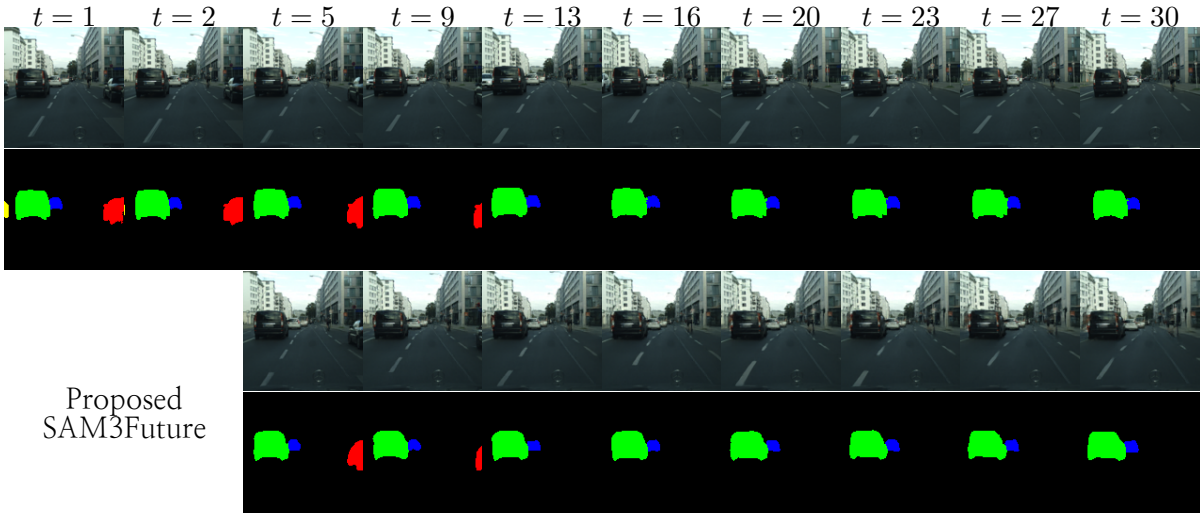


Figure 5: SAM3Future predictions on Cityscapes dataset (RGB + Object segmentation).

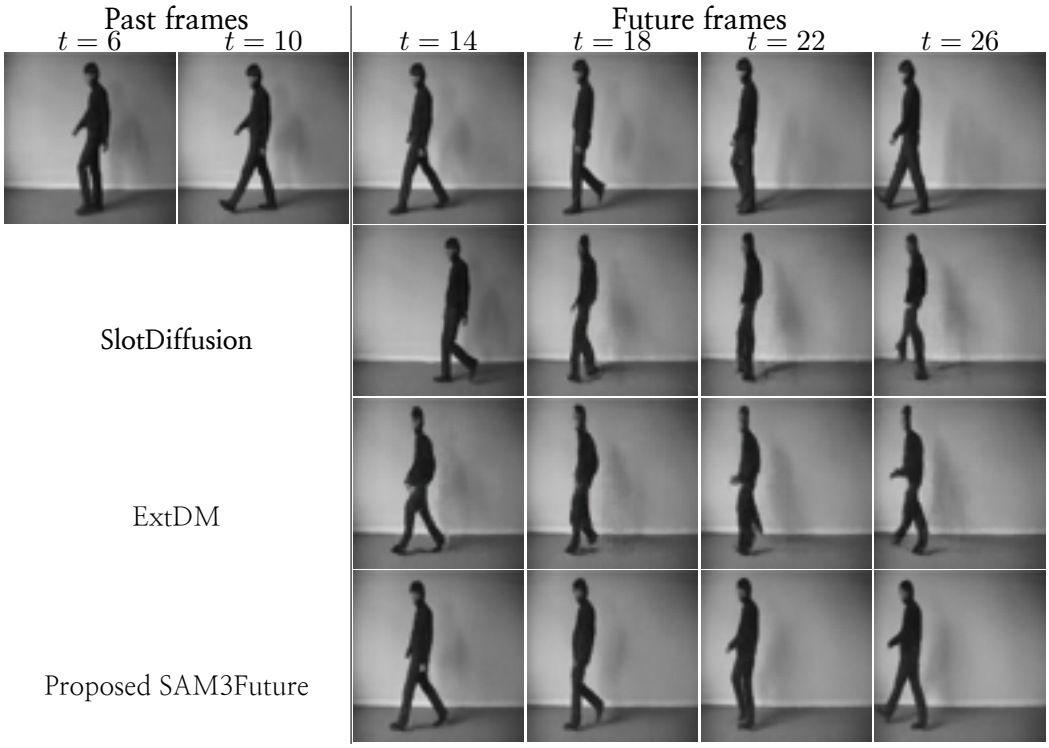


Figure 6: Qualitative comparisons between different video prediction models Cityscapes dataset.

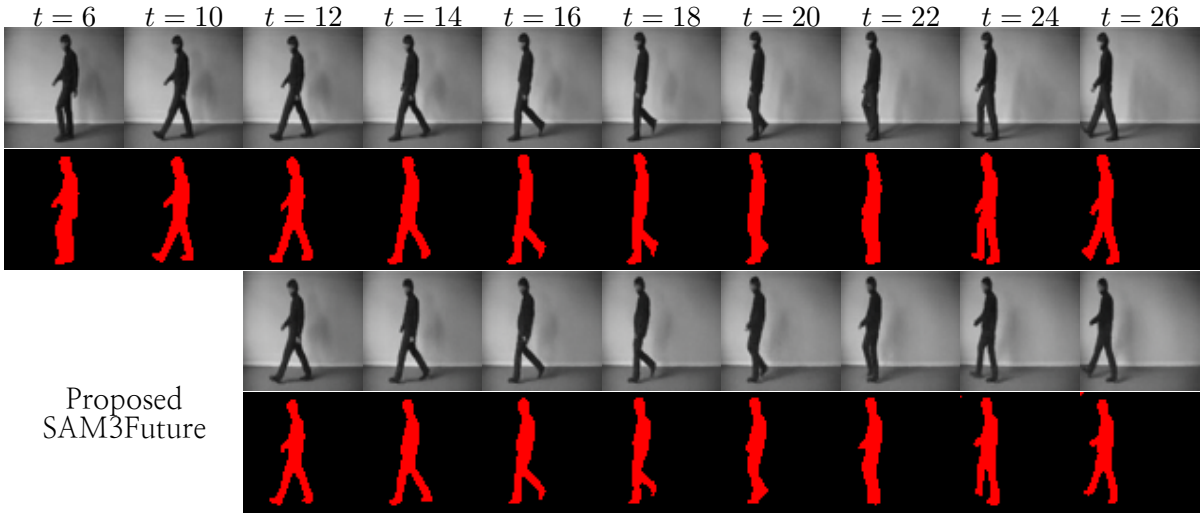


Figure 7: SAM3Future predictions on KTH dataset (RGB + Object segmentation).

Method	PSNR ↑	SSIM ↑	LPIPS ↓
OCVP[15]	31.20	0.925	0.135
SlotDiffusion[14]	31.30	<u>0.926</u>	0.089
LPWM[16]	<u>31.45</u>	0.925	<u>0.081</u>
SAM3Future	32.25	0.927	0.075

Table 3: Quantitative comparison on Obj3D dataset with OCVP methods.

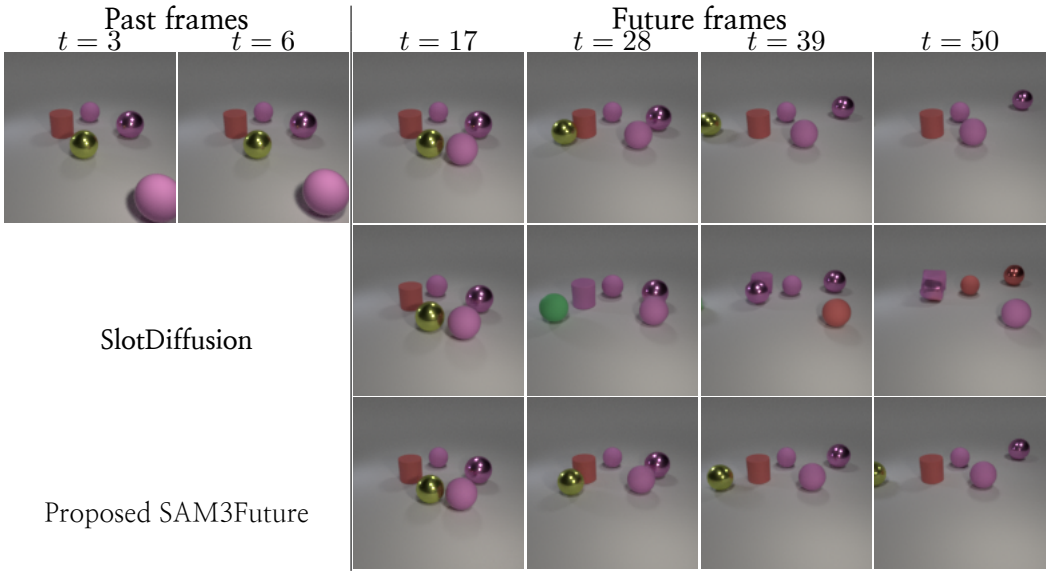


Figure 8: Qualitative comparisons between different video prediction models Obj3D dataset.

5.4. Comparisons between different video prediction methods

Figure 4,6,8 and Table1,2, 3 present both qualitative and quantitative comparisons of the proposed SAM3Future video prediction framework compared with SOTA video prediction methods. Overall, the proposed SAM3Future framework consistently outperforms SOTA video prediction methods on the Cityscapes, KTH, and Obj3D datasets.

Comparisons between different future conditions in diffusion-based methods

Table 1 and 2 show that our proposed method outperforms four SOTA video prediction methods on both datasets. For qualitative comparison, we compare our method with ExtDM[12], which achieves the second-best performance among the compared methods. Figure 4 and 6 show that both our method and ExtDM generate visually comparable background regions. In contrast, our approach better captures foreground object appearance across frames. In particular, our method reduce the problem where foreground objects are overly smoothed into the background over time. These results implies that our superior performance stems from our object-centric video representation conditioning strategy and the effectiveness of our method.

Comparisons between different object-centric video prediction methods

Table 3 and Figure 4,6,8 show that the proposed method outperforms three SOTA object-centric video prediction methods on Obj3D. For qualitative comparison, since SlotDiffusion[14] and LPWM[16] showed comparable performance, we visually compared with SlotDiffusion that employed diffusion process to video prediction. As shown in Figure 4, 6 and 8, SlotDiffusion often produces degraded object appearance and less stable

background structures over time. In contrast, our method maintains clearer foreground objects while preserving high-fidelity background regions across frames.

These results suggest that SAM3-derived object representations offer effective object-level guidance for future prediction by preserving consistent object identity across frames. Therefore, SAM3Future robustly handles complex object dynamics and camera motions in object-centric video prediction.

FOP	SSC	PSNR ↑	SSIM ↑	LPIPS ↓	FVD ↓
–	–	23.52	0.713	0.090	85.56
✓	–	23.61	0.718	0.091	84.53
✓	✓	24.31	0.747	0.082	78.41

Table 4: Module ablation in proposed SAM3Future on Cityscapes dataset.

5.5. Comparisons between different SAM3Future variants

This section investigates how each module design affects the performance of SAM3Future on the Cityscapes dataset. We evaluate the two key components of SAM3Future: the Future Object Predictor (FOP) and the Step-wise Spatial Conditioning (SSC) module.

Comparing the results in the first and second rows in Table 4 shows that the predicted future object embeddings derived from SAM3 serve as useful conditioning signal for the diffusion framework and improved the overall prediction quality. This improvement indicates that SAM3-derived object representations provide useful object-level guidance for modeling future dynamics, especially for foreground objects whose appearance and location change over time.

Comparing the second and fourth rows in Table 4 further demonstrates the effectiveness of Step-wise Spatial Conditioning module. This step-wise conditioning helps mitigate errors in the predicted pseudo cues and provides more reliable object-centric guidance during generation, leading to more accurate video predictions.

Consequently, the full model incorporating both components yields the best performance (fourth row in Table 4), confirming that both modules are effective and complementary.

6. Conclusion

In this work, we effectively leverage the powerful object-level representation capabilities of SAM3, a unified vision foundation model. SAM3 object representations provide instance-discriminative object-level guidance, which is effective for modeling foreground object in future frames. In addition, the proposed step-wise spatial conditioning dynamically localizes predicted future object embeddings during denoising, enabling adaptive conditioning throughout the generation process. By supervising the object-wise correlation maps with SAM3-derived masks, the proposed conditioning module encourages predicted object conditions to be spatially aligned with their corresponding object regions.

Experiments on the Cityscapes, KTH, and Obj3D datasets demonstrate that SAM3Future outperforms several SOTA video prediction methods, especially in preserving object-level dynamics and foreground appearance across frames.

Despite these achievements, proposed future object embedding predictor is trained implicitly through the denoising process. Therefore, explicitly modeling future object-level temporal changes remains an important direction for further improvement.

As future work, we aim to introduce explicit future object embedding prediction objectives to provide richer conditioning signals in terms of object-level temporal transitions.

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논문요약

Improving video prediction with SAM3 object-centric video representation

우한봄
전자전기컴퓨터공학과
성균관대학교

비디오 예측은 과거 관측 프레임으로부터 시간적 동역학을 모델링하여 미래 프레임을 예측하는 것을 목표로 한다. 최근 확산 기반 비디오 예측 방법들은 미래 프레임 생성에서 유망한 성능을 보여주었다. 불확실한 미래 동역학에 대한 명시적인 지침을 제공하기 위해, 기존 방법들은 주로 모션 표현 또는 객체 표현과 같은 예측된 미래 단서를 생성 과정의 조건으로 사용한다. 객체 표현은 비디오를 객체 수준에서 표현함으로써 구조화된 지침을 제공한다. 그러나 예측된 미래 객체 표현이 프레임 간 일관된 객체 정체성을 유지하지 못하는 경우, 예측 비디오에서 객체 경계가 과도하게 부드러워지는 문제를 여전히 방지하기 어렵다. 또한 확산 기반 비디오 예측에서 사용되는 고정 조건화 전략은 이러한 예측 단서의 오류를 생성 과정 전반에 걸쳐 샘플링 편향으로 반복적으로 전파할 수 있다.

이러한 문제를 해결하기 위해, 우리는 Segment Anything 3 (SAM3) 에서 얻은 정체성 일관 객체 비디오 표현을 미래 비디오 예측에 활용하는 조건부 확산 프레임워크인 SAM3Future 를 제안한다. SAM3Future 는 객체별 시간적 동역학을 모델링하여 미래 SAM3 객체 임베딩을 예측하고, 이를 사용해 객체 중심 비디오 표현을 구성한다. 불완전한 미래 예측의 영향을 완화하기 위해, 우리는 각 노이즈 제거 단계에서 예측된 객체 임베딩을 적응적으로 위치화하는 단계별 조건화 메커니즘을 도입한다. 세 가지 비디오 예측 벤치마크에서의 실험 결과는 SAM3Future 가 여러 비디오 예측 방법들을 능가함을 보여준다.

Keywords: 비디오 예측, 조건부 확산 모델, 객체 중심 표현, 노이즈 단계별 조건화